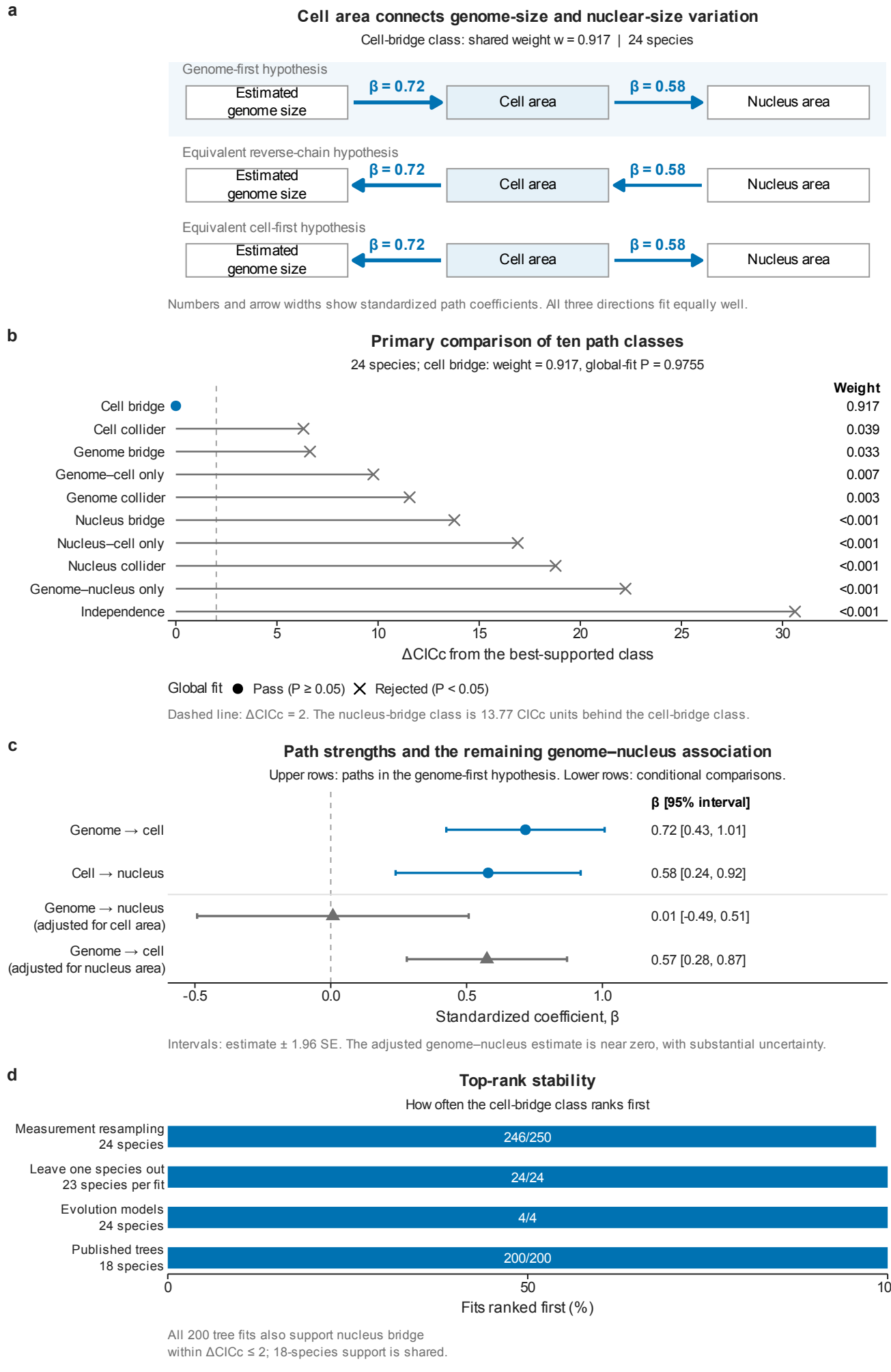
a
Cell area connects genome-size and nuclear-size variation
Cell-bridge class: shared weight w = 0.917 | 24 species
Genome-first hypothesis
Estimated genome size
β = 0.72
Cell area
β = 0.58
Nucleus area
Equivalent reverse-chain hypothesis
Estimated genome size
β = 0.72
Cell area
β = 0.58
Nucleus area
Equivalent cell-first hypothesis
Estimated genome size
β = 0.72
Cell area
β = 0.58
Nucleus area
Numbers and arrow widths show standardized path coefficients. All three directions fit equally well.
b
Primary comparison of ten path classes
24 species; cell bridge: weight = 0.917, global-fit P = 0.9755
Weight
Cell bridge 0.917
Cell collider 0.039
Genome bridge 0.033
Genome–cell only 0.007
Genome collider 0.003
Nucleus bridge <0.001
Nucleus–cell only <0.001
Nucleus collider <0.001
Genome–nucleus only <0.001
Independence <0.001
0
5
10
15
20
25
30
ΔCICc from the best-supported class
Global fit
Pass (P ≥ 0.05)
Rejected (P < 0.05)
Dashed line: ΔCICc = 2. The nucleus-bridge class is 13.77 CICc units behind the cell-bridge class.
c
Path strengths and the remaining genome–nucleus association
Upper rows: paths in the genome-first hypothesis. Lower rows: conditional comparisons.
β [95% interval]
Genome → cell
0.72 [0.43, 1.01]
Cell → nucleus
0.58 [0.24, 0.92]
Genome → nucleus (adjusted for cell area)
0.01 [-0.49, 0.51]
Genome → cell (adjusted for nucleus area)
0.57 [0.28, 0.87]
-0.5
0.0
0.5
1.0
Standardized coefficient, β
Intervals: estimate ± 1.96 SE. The adjusted genome–nucleus estimate is near zero, with substantial uncertainty.
d
Top-rank stability
How often the cell-bridge class ranks first
Measurement resampling 24 species
246/250
Leave one species out 23 species per fit
24/24
Evolution models 24 species
4/4
Published trees 18 species
200/200
0
50
100
Fits ranked first (%)
All 200 tree fits also support nucleus bridge within ΔCICc ≤ 2; 18-species support is shared.